



H2S

BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : SSR0

Team Leader Name :PATADIYA SMIT

Problem Statement :3

Development of Surface AQI & Identification of HCHO Hotspots over India using Satellite Data

Team Members & Responsibilities

Equal Contribution • Complementary Expertise • Shared Ownership

TEAM LEADER

PATADIYA SMIT

Project Lead & ML Lead



Vidush Somany Institute of
Technology and Research



TEAM MEMBER 1

SONI SHUBH

Data Science & Analytics Lead



SAL College of Engineering



KEY TASKS & RESPONSIBILITIES

01



Data Collection & Pipeline

Acquire and integrate Sentinel-5P, CPCB, FIRMS, ERA5 and auxiliary datasets.

02



Data Preprocessing & Cleaning

Harmonization, quality control, gap-filling, feature engineering.

03



Model Development

Design and train CNN-LSTM + Attention model for AQI prediction.

04



Model Evaluation & Validation

Run experiments, compute metrics, perform three validation protocols.

05



Explainability & Interpretability

SHAP analysis, feature importance, and insight generation.

06



Documentation & Reporting

Prepare methodology, results, charts, and final submission report.

KEY TASKS & RESPONSIBILITIES

01



HCHO Hotspot Detection

Process TROPOMI HCHO data and detect hotspots using DBSCAN clustering.

02



Fire Correlation Analysis

Correlate HCHO hotspots with FIRMS fire data and analyze spatial patterns.

03



Wind Transport Analysis

Compute wind vectors and analyze transport patterns using ERA5 data.

04



Dashboard & Visualization

Develop interactive dashboard (Streamlit/Plotly) for AQI & hotspot visualization.

05



Database & Backend Management

Manage data storage, queries, and performance optimization.

06



Presentation & Communication

Create slides, present findings, handle Q&A, and ensure cohesive communication.



Together We Deliver

From satellite data to actionable insights for cleaner air, smarter decisions, and healthier cities.



Our Approach

Collaborate equally • Communicate effectively • Build impact together

The Invisible Crisis — & Why Satellites Must Solve It

Research-backed context • Problem Statement 3 • ISRO BAH 2026

2 Million

premature deaths/year

India — air pollution is the #1 cause of death

Lancet 2024

64%

districts — zero monitor

Of India's 742 districts, no real-time AQI data

CSE India 2025

1.4 Billion

people above WHO limit

Entire India exposed to $PM_{2.5} > 10\times$ WHO guideline

Lancet Planetary Health

+1.70%

HCHO rise per year

Increasing HCHO trend over India (TROPOMI data)

ACP Journal 2024



INSAT-3D AOD $\rightarrow PM_{2.5}$

$R^2 > 0.75$ correlation. Pre-monsoon $PM_{2.5}$ hits $103 \mu g/m^3$ — $20\times$ WHO limit. Statistical models alone insufficient for full India coverage.

ScienceDirect | Atmos. Env. 2020



CNN-LSTM AQI Forecasting

Hybrid CNN-LSTM outperforms LSTM-only by 23% RMSE. Spatial + temporal fusion critical. No study uses INSAT-3D + TROPOMI + CPCB jointly.

Springer Nature | Disc. App. Sci. 2025



TROPOMI HCHO + Biomass Burning

Punjab post-monsoon: $34,000 \text{ km}^2$ burnt (Sentinel-2). HCHO $+1.7\%/yr$. Fire-to-HCHO transport not yet spatially mapped over full India.

ACP Journal | Atmos. Chem. Phys. 2024

⚠️ Current reality: static ground stations • single-pollutant • no spatial coverage • no INSAT-3D DL fusion



✅ SSRO: INSAT-3D + TROPOMI + ERA5 + CPCB • CNN-LSTM • NAAQs AQI • HCHO hotspot maps • daily India-wide

Integrated AI-based Surface AQI & HCHO Monitoring System

91 – Roger

Three validation methods, reported honestly

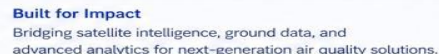
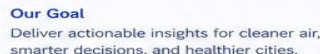
the absolutely worst of these, holding the score that corresponds to the lowest value of the Area Under the Curve (AUC) for the test data

OPTIMISTIC - LEAKAGE PRESENT		BLEST METHODOLOGICALLY FAIR		NAÏVETÉ - REAL DISTRIBUTION SHOWN	
Random data-window split		Mixed effect (strata = 5)		Temporal holdout (Out-Of-Sequence)	
AUC	0.918 / 0.932	AUC	0.475 / 0.472	AUC	0.228 / 0.229
RMSE	54.19	RMSE	58.83	RMSE	57.49
RMSE	58.84	RMSE	64.45	RMSE	67.41
Category AUC	71.87	Category AUC	58.76	Category AUC	67.47
Temp. AUC	59.85	Temp. AUC	58.76	Temp. AUC	67.41

Reduce the test set information, so that you end up with data that is different from what you trained on. This is the worst of the three validation methods - it doesn't reflect reality, so it's biased

Reduce realism using nested validation. This is the best method, as it reflects the way in which you would actually use the model in practice

Reduce realism using nested validation. This is the best method, as it reflects the way in which you would actually use the model in practice



PROCESS FLOW DIAGRAM – TEAM SSRO

Integrated AI, Remote Sensing & Geospatial Intelligence Platform

1 MULTI-SOURCE DATA ACQUISITION



TROPOMI S5P
NO₂, HCHO, CO,
O₃, SO₂
(L3 Products)



CPCB CAAQMS
Ground AQI Data
5 Cities
(2024 Daily)



ERA5-LAND
Wind, Temp.,
Humidity, PBL,
Pressure



FIRMS
Fire Data
MODIS + VIIRS
(Oct–Nov 2024)



PREPROCESSING

- Quality Assurance
- Spatial & Temporal Collocation
- Regridding (0.1° × 0.1°)
- Missing Value Handling
- Feature Engineering (Lags, Rolling, Ratios)

2 OBJECTIVE 1 – SURFACE AQI PREDICTION



MODEL BENCHMARKING & SELECTION

- Linear Regression
- Ridge Regression
- Random Forest
- Gradient Boosting
- XGBoost
- Support Vector Regression
- MLP (Neural Network)
- CNN-LSTM
- CNN-LSTM + Attention

8 Models Evaluated



BEST MODEL SELECTION

Best Performing Model Selected
(Based on CV & Test Performance)



EXPLAINABLE AI

- SHAP (Feature Importance)
- Attention Visualization
- Feature Importance Plots



PREDICTION CONFIDENCE ESTIMATION

Prediction Confidence Estimation
(Monte Carlo Dropout)



SURFACE AQI MAPS

Daily Surface AQI Maps over India (0.1° Resolution)
132-point Delhi Grid

3 OBJECTIVE 2 – HCHO HOTSPOTS & SOURCE ATTRIBUTION



TROPOMI HCHO (L3)
HCHO Columns (0.1° Grid)
567 Grid Points



FIRE DATA (FIRMS)
MODIS + VIIRS Active Fire
Oct–Nov 2024 (73 Fire Points)



ERA5 WIND FIELDS
U/V Wind Vectors
Transport Conditions



ADAPTIVE HOTSPOT DETECTION
95th Percentile Thresholding + DBSCAN
(2 Clusters Found)



FIRE-HCHO CORRELATION
Correlation Analysis
 $r = -0.24$ (Transport Signal)



SOURCE ATTRIBUTION
Fire Source → Meteorological Transport → Downwind HCHO Enhancement
Example: Punjab Fires → NW Winds → Delhi HCHO Enhancement

4 INTEGRATED OUTPUTS & DECISION SUPPORT



SURFACE AQI MAPS

Daily AQI at 0.1° Resolution Across India



HCHO HOTSPOT MAPS

High-resolution Hotspot Maps (567 Grid Points)



FIRE-HCHO CORRELATION

Quantitative Relation Between Fires & HCHO Enhancement



WIND TRANSPORT ANALYSIS

Wind Pathways & Pollution Transport Insights



EXPLAINABILITY REPORTS

SHAP, Attention & Feature Importance Reports



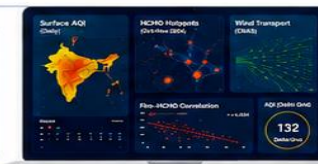
VALIDATION METRICS

RMSE, MAE, R², Pearson Correlation, Confidence Scores



INTERACTIVE DASHBOARD

GIS Visualization, Analytics, Download & Reporting



QUICK WORKFLOW

1



DATA COLLECTION
Satellite, Ground & Meteorological Data

2



DATA PROCESSING
Harmonization, Collocation & Feature Engineering

3



AI ANALYSIS
Model Benchmarking, Prediction & Explainability

4



SCIENTIFIC INSIGHTS
AQI Prediction, HCHO Hotspots, Correlation & Source Attribution

5



DECISION SUPPORT
Interactive Maps, Reports & Actionable Intelligence



Dual-objective Integrated Platform



Multi-source Satellite + Ground Data Fusion



AI-driven Prediction & Hotspot Intelligence



Explainable, Validated & Reliable Results



Actionable Insights for Policy & Decision Makers

SOLUTION WIREFRAME

Interactive Decision Support Dashboard (Proof of Concept) - Satellite-to-Surface AQI Platform

AI-powered dashboard demonstrating Surface AQI prediction, HCHO hotspot intelligence, explainable AI and scientific validation using a five-city pilot dataset.

OBJECTIVE 1

Satellite-to-Surface AQI

Predict city-level AQI using Sentinel-5P TROPOMI + ML (CNN-LSTM + Attention)

OBJECTIVE 2

HCHO Hotspots, Fire & Wind Transport

Detect HCHO hotspots, correlate with fires, and analyze wind transport patterns

PROOF OF CONCEPT

5 CPCB Cities
2024 Pilot Dataset

Overview

Pilot Dataset

AQI Explorer

Model Benchmark

Explainable AI (SHAP)

HCHO Analysis

Wind Transport

Validation

Reports

Data Status

Live

Last Updated:
02 Dec 2024, 10:30 AM

PILOT DATASET

5 CPCB Cities

Delhi, Mumbai, Chennai,
Bengaluru, Hyderabad

MATCHED CITY-DAY RECORDS

1,473

Real city-day
observations

VALIDATION SCORE (R^2 / R)

0.92 / 0.47

Optimistic / Rigorously
validated

SELECTED MODEL

CNN-LSTM +
Attention

$R^2 = 0.875$ | Acc = 71.3%

STUDY PERIOD

Jan-Dec 2024

Full year TROPOMI x
CPCB matched

RESEARCH WORKFLOW (PROOF OF CONCEPT)

- 1. Data Collection**
Sentinel-5P, CPCB,
FIRMS, ERA5, Weather
- 2. Processing**
Harmonization, QC,
Gap-filling, Alignment
- 3. AI Prediction**
CNN-LSTM + Attention
(Surface AQI / HCHO)
- 4. Explainability**
SHAP Attributions,
Feature Insights
- 5. Decision Support**
Dashboards, Alerts,
Reports & Insights

01 Pilot Dataset

02 Model

03 Benchmark

04 SHAP

05 Validation

06 Objective 2

07 Generalization Test

Pilot AQI Explorer (Delhi Demonstration)

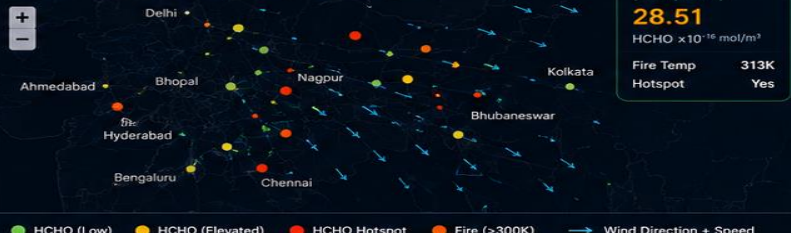


MEAN AQI
218

MAX AQI
494

OBSERVATION PERIOD
313 Days

Representative HCHO Hotspot Analysis (Pilot Region)



Benchmark Results (Pilot Evaluation)

RANK	MODEL	R^2	CATEGORY ACC.
01	Gradient Boosting	0.875	71.3%
02	Random Forest	0.871	70.0%
03	XGBoost	0.842	70.0%
04	Neural Net (MLP)	0.790	62.0%
05	SVR	0.723	60.2%
06	MLP	0.712	59.1%
07	CNN-LSTM + Attention	0.701	61.5%

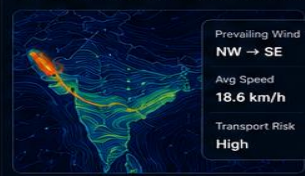
SHAP Explainability (Selected Model)



Validation Protocols (Three Methods)

Random Split (Optimistic)	Blocked Spatial Split (Stride-7)	Temporal Holdout (Last 30 Days)
R^2 / R 0.918 / 0.958	R^2 / R 0.472 / 0.695	R^2 / R 0.228 / 0.597
MAE: 14.19 RMSE: 20.84 Acc: 77.6%	MAE: 29.52 RMSE: 44.46 Acc: 55.1%	MAE: 54.49 RMSE: 87.41 Acc: 49%
Tests: 340	Tests: 49	Tests: 447

Transport Pathway Analysis (Pilot Demo)



Pilot Dataset

TROPOMI • CPCB (5 Cities)
FIRMS • ERA5 • Weather



AI Prediction

CNN-LSTM + Attention
Surface AQI & HCHO



Scientific Validation

Multi-protocol validation
(Robust & Generalizable)



Explainable AI

SHAP • Feature Insights
Transparency

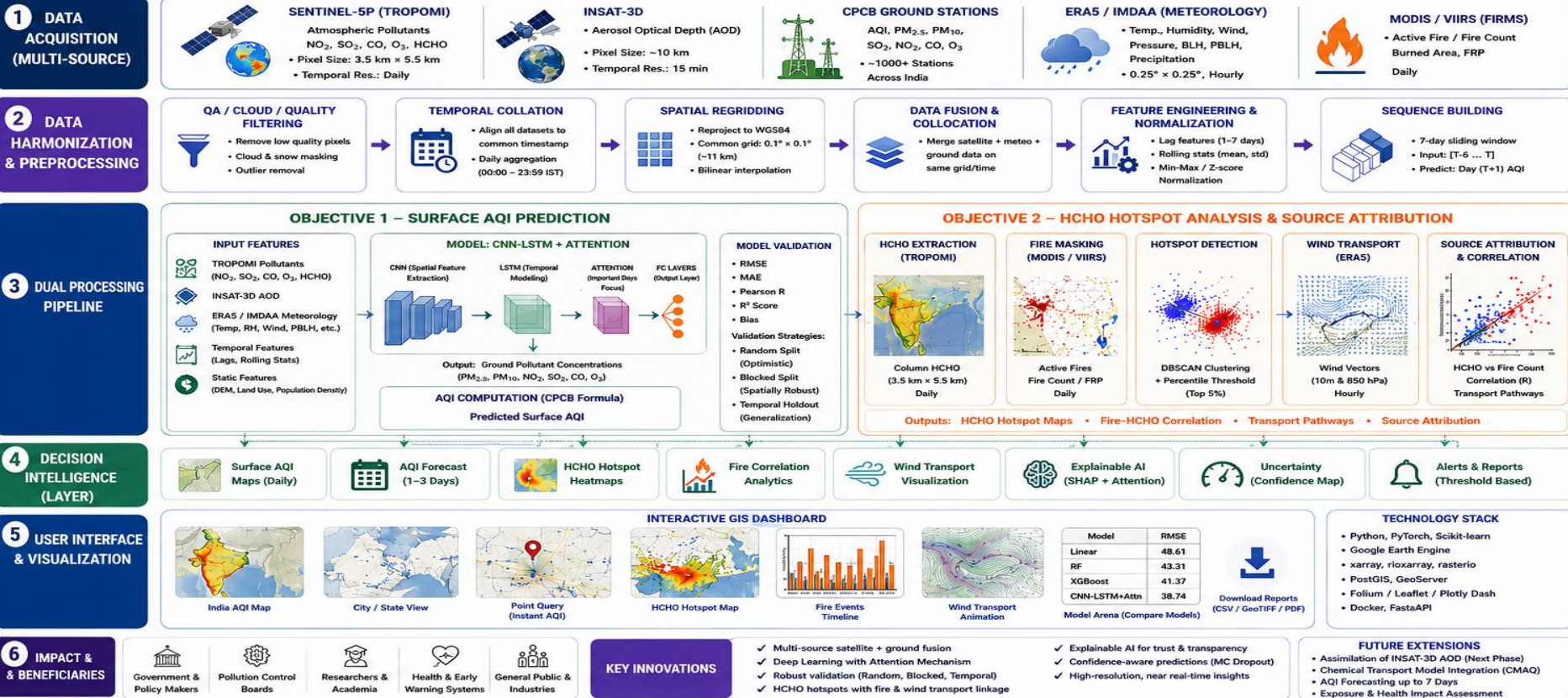


Decision Support Prototype

Dashboards • Alerts
Insights & Reports

AI-DRIVEN SATELLITE-BASED SURFACE AQI PREDICTION & HCHO HOTSPOT ANALYSIS FRAMEWORK

Integrating Multi-Source Earth Observation, AI/ML & Geospatial Analytics for Environmental Intelligence



TECHNOLOGY STACK

Integrated AI, Remote Sensing and Geospatial Computing Ecosystem

1

AI & MACHINE LEARNING



FRAMEWORKS

- Python 3.11
- PyTorch 2.x
- PyTorch Lightning
- scikit-learn

MODELS EVALUATED (MODEL ARENA)

- Linear Regression
- Ridge Regression
- Random Forest
- Gradient Boosting
- XGBoost
- Support Vector Regression (SVR)
- Feed-forward Neural Network (MLP)
- CNN-LSTM
- CNN-LSTM + Attention

(Final Selected Model)



2

EARTH OBSERVATION DATA SOURCES



- Sentinel-5P TROPOMI L3
 - NO₂, SO₂, CO, O₃, HCHO
- INSAT-3D AOD
(planned integration)
- MODIS Active Fire (FIRMS)
- VIIRS Active Fire (FIRMS)
- CPCB CAAQM Stations
- ERA5 / IMDAA Meteorology



3

GEOSPATIAL ANALYTICS



- Google Earth Engine
- GeoPandas
- Cartopy
- Rasterio
- Xarray
- Rioxarray
- Shapely
- Folium



4

SCIENTIFIC COMPUTING



- NumPy
- Pandas
- NetCDF4
- h5py
- SciPy
- Dask
- Joblib



5

EXPLAINABLE AI & VALIDATION



- SHAP (Feature Importance)
- Attention Visualization
- Feature Importance
- RMSE (Root Mean Square Error)
- MAE (Mean Absolute Error)
- R² (Coefficient of Determination)
- Pearson Correlation
- Temporal Hold-out Validation



6

VISUALIZATION & DEPLOYMENT

VISUALIZATION



DASHBOARD / FRONTEND



DEPLOYMENT & COLLABORATION



FINAL SELECTED MODEL

CNN-LSTM + ATTENTION

MODEL ARENA (COMPREHENSIVE EVALUATION)

Linear | Ridge | Random Forest | Gradient Boosting | XGBoost | SVR | MLP | CNN-LSTM | CNN-LSTM + Attention

Estimated Implementation Cost (Proof of Concept)

Cost breakdown for building our Satellite-to-Surface AQI & HCHO Intelligence Platform using a five-city pilot dataset.



PROOF OF CONCEPT

5 CPCB Cities
2024 Pilot Dataset

1



Phase 1 – Data Acquisition & Engineering

- Sentinel-5P (TROPOMI) via GEE
- CPCB Air Quality Data (5 Cities)
- ERA5 / IMDAA Meteorology
- MODIS / VIIRS (FIRMS) Fire Data
- Data Harmonization, QC & Cleaning
- Feature Engineering

Infrastructure / Tools



ESTIMATED COST

₹ 0

(Free Academic / Open Data)

2



Phase 2 – AI Model Development

- Model Benchmarking: Linear, Ridge, RF, GBM, XGBoost, SVR, MLP
- Deep Learning: CNN-LSTM, CNN-LSTM + Attention
- Hyperparameter Tuning
- SHAP Explainability
- Validation (RMSE, MAE, R^2 , etc.)

Infrastructure / Tools



ESTIMATED COST

₹ 2,000 – ₹ 6,000

(GPU Compute for Training)

3



Phase 3 – Interactive Dashboard & GIS

- AQI Prediction Dashboard
- HCHO Hotspot & Fire Analytics
- Wind Transport Visualization
- Maps, Charts, Analytics
- Reports & Data Download

Infrastructure / Tools



ESTIMATED COST

₹ 0

(Open-Source Stack)

4



Phase 4 – Scientific Valisboard & Reporting

- Statistical Validation (RMSE, MAE, R^2)
- Correlation & Significance Analysis
- Visual Analysis & Figures
- Scientific Report & Documentation

Infrastructure / Tools



ESTIMATED COST

₹ 0

(Open-Source Tools)

5



Phase 5 – Prototype Deployment

- Web Hosting (Dashboard)
- Code Repository & Version Control
- Public Demo & Access
- CI/CD (Optional)

Infrastructure / Tools



ESTIMATED COST

₹ 0

(Free Tier Hosting)



WHAT IS INCLUDED

- ✓ Pilot dataset: 5 CPCB cities
- ✓ Satellite & ground data
- ✓ AI model benchmarking
- ✓ Explainable AI (SHAP)
- ✓ Scientific validation
- ✓ Interactive dashboards
- ✓ HCHO & fire intelligence
- ✓ Wind transport analysis

FUTURE (OPTIONAL) OPERATIONAL DEPLOYMENT



If operationalized with automated pipeline, API, monitoring & cloud infrastructure.

Estimated:

₹ 5,000 – ₹ 13,000 / month

(Not part of hackathon scope)



TOTAL HACKATHON IMPLEMENTATION COST (PHASE 1–5)

₹ 2,000 – ₹ 6,000



KEY TAKEAWAY

Our solution leverages public satellite data, government datasets and open-source technologies to build a scalable and cost-effective platform.



Designed for low-cost scalability. Can be extended to national level with minimal incremental cost.



All datasets are public or open-access. | All tools are open-source or academic. | Built for scalability, transparency & societal impact.



BHARATIYA
ANTARIKSH
HACKATHON

2026

THANK YOU

Team SSRO — Problem Statement 3

Surface AQI & HCHO Hotspot Identification Over India`

PATADIYA SMIT • SONI SHUBH